

```
import numpy as np
import pandas as pd
from collections import defaultdict
import re
```

```
df_fake = pd.read_csv('Fake.csv')
```

```
df_fake.head()
```

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
df_fake=df_fake.drop(['subject','date'],axis=1)
```

```
df_fake.head()
```

	title	text
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...

```
df_fake.text.shape
```

```
(23481,)
```

```
df_fake['label'] = 'FAKE'
```

```
df_fake.head()
```

	title	text	label
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	FAKE
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	FAKE
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	FAKE
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	FAKE
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	FAKE

```
df_true = pd.read_csv('True.csv')
```

```
df_true.head()
```

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017

```
df_true=df_true.drop(['subject','date'],axis=1)
```

```
df_true.head()
```



	title	text
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...

```
df_true['label']='TRUE'
df_true.head()
```



	title	text	label
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	TRUE
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	TRUE
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	TRUE
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	TRUE
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	TRUE

```
data_train = pd.concat([df_true, df_fake], ignore_index=True)
data_train.head()
```



	title	text	label
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	TRUE
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	TRUE
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	TRUE
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	TRUE
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	TRUE

```
data_train.shape
```



```
(44898, 3)
```

```
data_train.to_csv('fake_news.csv')
```

```
import pandas as pd
```

```
import numpy as np
```

```
import itertools
```

```
import seaborn as sns
```

```
import nltk, re, string
```

```
from string import punctuation
```

```
from nltk.corpus import stopwords
```

```
from sklearn import metrics
```

```
import matplotlib.pyplot as plt
```

```
from wordcloud import WordCloud, STOPWORDS
```

```
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, roc_auc_score
```

```
from sklearn.model_selection import train_test_split
```

```
from keras.models import Model
```

```
from keras.layers import Dense, Embedding, Input, LSTM, Bidirectional, GlobalMaxPool1D, Dropout
```

```
import tensorflow as tf
```

```
from tensorflow.keras.preprocessing.text import Tokenizer
```

```
#from keras.preprocessing.text import Tokenizer
```

```
from keras.preprocessing.sequence import pad_sequences
```

```
nlTK.download('stopwords')
```

```
[nlTK_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

```
df=pd.read_csv('fake_news.csv')
```

```
df.head()
```

```
↗
```

	Unnamed: 0		title		text	label
0	0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...		TRUE	
1	1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...		TRUE	
2	2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...		TRUE	
3	3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...		TRUE	
4	4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...		TRUE	

```
#removing news source info from text
```

```
def extract_txt(text):
```

```
    regex = re.search(r"(?<=(Reuters\)\s\-\s).*", text)
```

```
    if regex:
```

```
        return regex.group(0)
```

```
    return text
```

```
df['text processed']=df['text'].apply(extract_txt)
```

```
df.drop(['Unnamed: 0'], axis=1, inplace=True)
```

```
df["label"]=df.label.apply(lambda x:0 if x=='TRUE' else 1)
```

```
df.head()
```

```
↗
```

		title		text	label		text processed
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	0	The head of a conservative Republican faction ...			
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	0	Transgender people will be allowed for the fir...			
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	0	The special counsel investigation of links bet...			
3	FBI Russia probe helped by Australian	WASHINGTON (Reuters) - Trump campaign	0	Trump campaign adviser George			

```
df['final news']= df['title']+" "+df['text processed']
```

```
df.head()
```

```
↗
```

		title		text	label		text processed		final news
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	0	The head of a conservative Republican faction ...			As U.S. budget fight looms, Republicans flip t...		
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	0	Transgender people will be allowed for the fir...			U.S. military to accept transgender recruits o...		
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	0	The special counsel investigation of links bet...			Senior U.S. Republican senator: 'Let Mr. Muell...		
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	0	Trump campaign adviser George Papadopoulos tol...			FBI Russia probe helped by Australian diplomat...		
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	0	President Donald Trump called on the U.S. Post...			Trump wants Postal Service to charge 'much mor...		

```
cols_del = [ 'title', 'text', 'text processed']
```

```
df=df.drop(cols_del,axis=1)
```

```
df.head()
```

	label	final news
0	0	As U.S. budget fight looms, Republicans flip t...
1	0	U.S. military to accept transgender recruits o...
2	0	Senior U.S. Republican senator: 'Let Mr. Muell...
3	0	FBI Russia probe helped by Australian diplomat...
4	0	Trump wants Postal Service to charge 'much mor...

```
stop=stopwords.words('english')

def cleanText(txt):
    # lowercaing
    txt = txt.lower()

    # removing stopwords
    txt = ' '.join([word for word in txt.split() if word not in (stop)])

    # removing non-alphabetic characters
    txt = re.sub('[^a-z]', '', txt)

    return txt

df['final news'] = df['final news'].apply(cleanText)
df.head()
```

	label	final news
0	0	usbudgetfightloomsrepublicansflipfiscalscripth...
1	0	usmilitaryaccepttransgenderrecruitsmondaypenta...
2	0	seniorusrepublicansenatorletrmuellerjobspecia...
3	0	fbirussiaprobehelpeaustraliandiplomatpoffny...

```
#path='/content/drive/MyDrive'

EMBEDDING_FILE=f'/glove.68.50d.txt'

# configuration setting

MAX_SEQUENCE_LENGTH = 100

MAX_VOCAB_SIZE = 20000

EMBEDDING_DIM = 50

VALIDATION_SPLIT = 0.2

BATCH_SIZE = 32

EPOCHS = 10

y = df ['label'].values

X = df.drop(['label'], axis=1)

y.shape

(44898,)

tokenizer=Tokenizer(num_words=MAX_VOCAB_SIZE)
tokenizer.fit_on_texts (list(X['final news']))
X = tokenizer.texts_to_sequences (list(X['final news']))

X = pad_sequences (X, maxlen=MAX_SEQUENCE_LENGTH)
print('Shape of data tensor:', X.shape)
```

↗ Shape of data tensor: (44898, 100)

```
word2idx=tokenizer.word_index
print('Found %s unique tokens.' % len (word2idx))
```

↗ Found 39090 unique tokens.

```
# prepare embedding matrix
```

```
print('Filling pre-trained embeddings...')
```

```
num_words = min(MAX_VOCAB_SIZE, len (word2idx) + 1)
```

```
embedding_matrix = np.zeros((num_words, EMBEDDING_DIM))
```

```
for word, i in word2idx.items():
```

```
    if i < MAX_VOCAB_SIZE:
```

```
        embedding_vector = word2vec.get(word)
```

```
        if embedding_vector is not None:
```

```
# words not found in embedding index will be all zeros.
```

```
        embedding_matrix[i] = embedding_vector
```

↗ Filling pre-trained embeddings...

```
# load pre-trained word embeddings into an Embedding layer
```

```
# note that we set trainable embedding_layer
```

```
embedding_layer=Embedding(
```

```
num_words,
```

```
EMBEDDING_DIM,
```

```
weights=[embedding_matrix],
```

```
input_length=MAX_SEQUENCE_LENGTH,
```

```
trainable=False
```

```
)
```

↗ /usr/local/lib/python3.10/dist-packages/keras/src/layers/core/embedding.py:90: UserWarning: Argument `input\_length` is deprecated. : warnings.warn(

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, stratify=y, random_state=0)
```

```
print('Building model...')
```

```
# create an LSTM network with a single LSTM
```

```
input_=Input(shape=(MAX_SEQUENCE_LENGTH,))
```

```
x = embedding_layer(input_)
```

```
# x = LSTM(15, return_sequences=True)(x)
```

```
x = Bidirectional (LSTM (15, return_sequences=True))(x)
```

```
x = GlobalMaxPool1D()(x)
```

```
output = Dense (1, activation="sigmoid")(x)
```

```
model=Model(input_, output)
```

```
model.compile( loss='binary_crossentropy', optimizer='adam',
```

```
metrics=['accuracy']
```

```
)
```

```
model.summary()
```

Building model...
Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 100)	0
embedding (Embedding)	(None, 100, 50)	1,000,000
bidirectional (Bidirectional)	(None, 100, 30)	7,920
global_max_pooling1d (GlobalMaxPooling1D)	(None, 30)	0
dense (Dense)	(None, 1)	31

Total params: 1,007,951 (3.85 MB)
Trainable params: 7,951 (31.06 KB)
Non-trainable params: 1,000,000 (3.81 MB)

```
print("Training model....")
```

```
r=model.fit(X_train, y_train, batch_size=BATCH_SIZE, epochs=EPOCHS, validation_split=VALIDATION_SPLIT)
```

Training model....

```
Epoch 1/10
898/898 ————— 71s 75ms/step - accuracy: 0.5183 - loss: 0.6924 - val_accuracy: 0.5131 - val_loss: 0.6930
Epoch 2/10
898/898 ————— 83s 76ms/step - accuracy: 0.5270 - loss: 0.6918 - val_accuracy: 0.5131 - val_loss: 0.6929
Epoch 3/10
898/898 ————— 70s 78ms/step - accuracy: 0.5299 - loss: 0.6914 - val_accuracy: 0.5131 - val_loss: 0.6929
Epoch 4/10
898/898 ————— 78s 74ms/step - accuracy: 0.5244 - loss: 0.6920 - val_accuracy: 0.5131 - val_loss: 0.6930
Epoch 5/10
898/898 ————— 81s 73ms/step - accuracy: 0.5274 - loss: 0.6917 - val_accuracy: 0.5131 - val_loss: 0.6931
Epoch 6/10
898/898 ————— 85s 77ms/step - accuracy: 0.5286 - loss: 0.6916 - val_accuracy: 0.5131 - val_loss: 0.6931
Epoch 7/10
898/898 ————— 81s 76ms/step - accuracy: 0.5223 - loss: 0.6923 - val_accuracy: 0.5131 - val_loss: 0.6932
Epoch 8/10
898/898 ————— 79s 73ms/step - accuracy: 0.5270 - loss: 0.6918 - val_accuracy: 0.5131 - val_loss: 0.6932
Epoch 9/10
898/898 ————— 84s 75ms/step - accuracy: 0.5289 - loss: 0.6915 - val_accuracy: 0.5131 - val_loss: 0.6929
Epoch 10/10
898/898 ————— 81s 74ms/step - accuracy: 0.5308 - loss: 0.6914 - val_accuracy: 0.5131 - val_loss: 0.6929
```

```
acc = r.history['accuracy']
```

```
val_acc = r.history['val_accuracy']
```

```
loss=r.history['loss']
```

```
val_loss=r.history['val_loss']
```

```
epochs_range = range (1, len (r.epoch) + 1)
```

```
print("Accuracy of the model on Training Dataris", model.evaluate(X_train, y_train) [1]*100)
```

```
print("Accuracy of the model on Testing Data is - ", model.evaluate(X_test,y_test) [1]*100)
```

1123/1123 ————— 23s 21ms/step - accuracy: 0.5320 - loss: 0.6915
Accuracy of the model on Training Dataris 52.299684286117554
281/281 ————— 5s 18ms/step - accuracy: 0.5241 - loss: 0.6920
Accuracy of the model on Testing Data is - 52.293986082077026

```
pred = model.predict(X_test)
```

```
pred[:5]
```

281/281 ————— 5s 18ms/step
array([[0.5186862],
[0.5186862],
[0.5186862],
[0.5186862],
[0.5186862]], dtype=float32)

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from sklearn.metrics import confusion_matrix
```

```
pred = model.predict(X_test)
pred[:5]
```

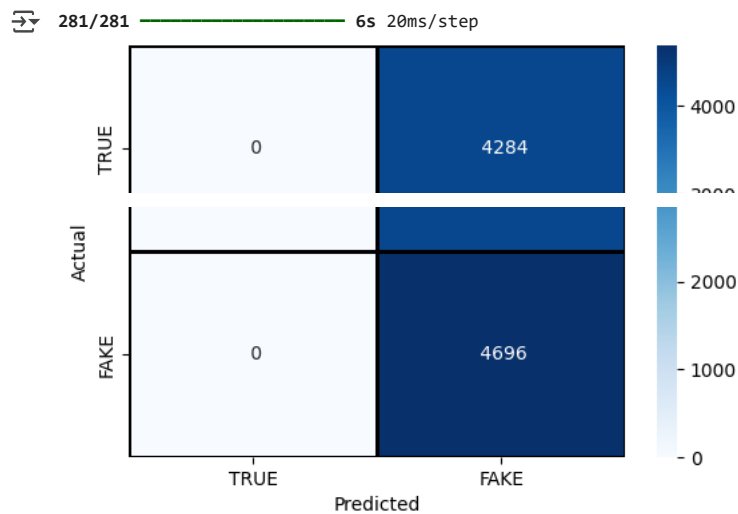
```

cm=confusion_matrix(y_test, pred.round())
cm=pd.DataFrame(cm, index=["TRUE", "FAKE"], columns=['TRUE', 'FAKE'])

# The following line was changed to include figsize as a parameter within plt.figure()
plt.figure(figsize=(6,4))

sns.heatmap(cm, cmap= "Blues", linecolor="black", linewidth = 1, annot=True, fmt=' ', xticklabels=["TRUE", "FAKE"], yticklabels=["TRUE", "FAKE"],
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()

```



```
y_pred=model.predict(X_test).ravel()
```

281/281 ————— 12s 42ms/step

```

testsent=["Trey Gowdy destroys this clueless DHS employee when asking about the due process of getting on the terror watch list. Her re
"Poland's new prime minister faces a difficult balancing act trying to repair bruised relations with the European imion withou

```

```

def cleanText(t):
    t = t.lower()
    t = ' '.join([word for word in t.split() if word not in (stop)])
    t = re.sub('[^a-z]', '', t)
    return t

import pandas as pd
import re
from keras.preprocessing.sequence import pad_sequences

def predict_text(lst_text):
    # Tokenize the input text
    test = tokenizer.texts_to_sequences(lst_text)

    # Pad the sequences to ensure consistent length
    test = pad_sequences(test, maxlen=MAX_SEQUENCE_LENGTH) # Fixed: pad_sequences should be applied to 'test'

    # Create a DataFrame with the original text for display later
    df_test = pd.DataFrame(lst_text, columns=['test_sent']) # Fixed: using lst_text instead of 1st_text

    # Predict sentiment using the model
    prediction = model.predict(test)

    # Add prediction results to the DataFrame
    df_test['prediction'] = prediction

    # Clean the text for better readability in the output
    df_test["test_sent"] = df_test["test_sent"].apply(cleanText)

    # Convert numerical predictions to categorical labels
    df_test['prediction'] = df_test['prediction'].apply(lambda x: "Fake" if x >= 0.5 else "Real")

    # Return the DataFrame with predictions
    return df_test
df_testsent=predict_text(testsent)
df_testsent

```

 1/1  0s 143ms/step

	test_sent	prediction
0	treygowdydestroyscluelessdhsemployeeaskingduep...	Fake
1	polandsnewprimeministerfacesdifficultbalancing...	Fake